

AST 325/326 (Fall 2026)

Lab #2 “Introduction to Dispersive Spectroscopy and Observations”

Lab report due: **Nov 9, 2026, 11:59 pm**, electronically on Quercus.

1. Overview

Spectroscopy is a fundamental tool used by all physical sciences. For astrophysicists, spectroscopy is essential for characterizing the physical nature of celestial objects and the universe. Astronomical spectroscopy has been used to measure the chemical composition and physical conditions (temperature, pressure, and magnetic field strength) in planets, stars, and galaxies, as well as their velocities. Characterizing a spectrograph’s instrumental parameters is the key to deriving the intrinsic spectrum of a source. In this lab you will use a spectrograph to collect data from common light sources (room lights and gas discharge lamps), establish a wavelength scale, and (weather permitting) measure astronomical spectra of stars and planets at the campus observatory. In class we will explore the fundamentals of diffraction, dispersion elements, 2-D detectors, telescopes, and review the celestial coordinate systems.

1.1 Schedule

This is a 4-week lab running from Oct 6 to Nov 5. Your lab report is due on Nov 9 at 11:59 pm.

Read Section 3.1.1 before the first Thursday lab on Oct 8.

Starting Oct 8, weather permitting, we will obtain astronomical spectra at the campus observatory. Each group has 1.5 hr in total, and must book an observing slot as early as possible:

<https://docs.google.com/spreadsheets/d/16S7JMHX0GXw4437MUclGRGky6UoEXQZzKQWqldP67qY/edit?usp=sharing>. Book early: groups that are clouded out need free slots later in the run to move into, and those go quickly.

Suggested pacing, and one rule. Week 1: read 3.1.1 before Thursday, collect your spectrometer data (3.1.2) in the Thursday lab (at AB88!!), and analyze it (3.1.3-3.1.5) afterwards. You can collect the data again in the Week 2 Thursday lab if you did not finish in Week 1. Weeks 1-4: observe at the campus telescope (3.2.2) when your slot comes, then reduce your telescope data (3.2.3-3.2.6) and write the report. Observing starts in the first week, so some of you will be at the telescope (3.2.2) before you have analyzed anything. That is fine: the analysis can wait. What cannot wait is the preparation. **Before your observing night, every group must have read Section 3.2.1 and must arrive with an observing plan: the targets you intend to observe, why, whether each will be up at a useful altitude during your slot, and the exposure time you intend to use for each.** A group that turns up without one will spend the night deciding instead of observing.

1.2 Goals

Use a simple visible-light spectrometer to explore the spectra of laboratory sources. We will measure the wavelength calibration of the Ocean Optics spectrometer, which will introduce the concept of linear least squares. We will then learn how to plan and carry out observations at a telescope: how to choose targets, work out whether and when they are observable from a given site, and estimate the exposure time each one needs. At the campus observatory we take spectra of astronomical sources with the LISA spectrograph on the 10-inch telescope, and learn the basic steps of astronomical data reduction (dark subtraction, flat fielding, wavelength solution, and spectral extraction). Along the way you will see that the raw output of a slit spectrograph is an image, not a spectrum, and that turning one into the other is most of the work.

1.3 References

- Thursday Lab Instrument (Section 3.1): Red Tide USB-650 spectrometer manual (Ocean Optics): [Installation and Operation Instructions \(PDF\)](#)
- Campus Observatory Instrument (Section 3.2): LISA spectrograph user guide (Shelyak, ref. DC0012A): [DC0012A-LISA-User-Guide-En.pdf](#).
- NIST Atomic Spectra Database, line list form: physics.nist.gov/PhysRefData/ASD/lines_form.html
- Howell, S. B., [Handbook of CCD Astronomy](#), Cambridge University Press, Chapters 1-4.
- Chromey, F. R., [To Measure the Sky](#), Cambridge University Press -- in particular Chapter 6 (Astronomical Telescopes), Chapter 8 (Detectors), and Chapter 11 (Spectrometers).
- Lectures and Lab #1 on statistics, error analysis, and least-squares fitting.

1.4 Key Steps

The lab is broken down into several stages:

1. Learn to operate the USB-650 spectrometer using the OceanView software.
2. Save spectra and read them into Python for plotting and analysis.
3. Observe and compare spectra of common sources -- incandescent lamp, fluorescent strip light, gas discharge lamps, and sunlight.
4. Determine the wavelength calibration of the spectrometer, i.e. the mapping between pixel number and wavelength. Do this by measuring the centroids (pixel positions) of lines of known wavelength, then use linear least squares to fit a polynomial and derive the wavelength solution.
5. Identify the unknown gases in two further discharge lamps using your wavelength solution.
6. Use the LISA spectrograph at the campus observatory to collect spectra of targets such as bright stars, the Moon and planets.
7. Reduce the LISA spectra (dark subtraction, flat fielding, wavelength solution, and extraction of a 1-D spectrum) and compare and contrast them.
8. Write up your report.

2. Spectrographs

For much of twentieth-century astronomy the detector was a photographic plate. Photographic plates are still used in some spectroscopic applications. Modern instruments use a CCD (charge-coupled device) or a CMOS sensor, in which an electronic wafer builds up an electrical charge when light shines on it. This charge is later read out and measured by a computer. Most spectrometers are fundamentally simple in design: a thin beam of light passes through, or is reflected from, an element which spreads out, or disperses, the light according to wavelength. An easy way of visualizing this is to think of a prism, which spreads out light into all the colors of the rainbow. The dispersing element (a prism or a diffraction grating) sends the violet light in one direction, the yellow light in a slightly different direction, the red light in still a different direction, and so on. This dispersed, polychromatic light is then focused onto a surface which acts as a detector.

One point is worth making before you start, because it is the single biggest surprise for students on their first night at a telescope. The plots of spectra you see in papers and textbooks -- intensity against wavelength -- are a *reduced product*. What a slit spectrograph actually records is a two-dimensional image: one axis is the dispersion direction (wavelength), and the other is the spatial direction (position along the slit). The emission lines of a calibration lamp appear as a picket fence of lines running across the frame; a star appears as a single horizontal streak; the night sky appears as lines running the full height of the slit. Nothing in that image is labeled in wavelength, the lines are usually slightly tilted or curved, and the sky is sitting on top of your target. Getting from that image to a plot of flux against wavelength is most of the work of spectroscopy, and it is what Section 3.2 is about. The Ocean Optics spectrometer in Section 3.1 hides all of this from you, because its detector is a single row of pixels -- which is exactly why we start there.

3. Activities

This lab consists of **two** parts. The first is the calibration of a small classroom spectrograph, which you will then use to identify an unknown gas. The second involves using the telescope on top of MP to measure real spectra of astronomical sources. The telescope calibration is more involved and is best undertaken after completing the first part. If your group is scheduled to observe before you have finished it, collect the telescope data that night anyway and perform the analysis later -- **but read Section 3.2.1 first and bring your target plan; that part is not optional.**

3.1 Ocean Optics Spectrograph

3.1.1 Overview and preparation -- **read before the Thursday lab (note that it will be in AB88)**

The USB-650 “Red Tide” spectrometer is a simple optical instrument based on a diffraction grating and a one-dimensional CCD detector array, so a spectrum reads out as a simple list of numbers -- one per pixel. Light enters via a slit located at the bottom of a threaded receptacle, which can be used to

connect an optical fiber terminated with an SMA plug. The detector is a Sony ILX511 linear silicon CCD; the instrument responds from about 350 to 1000 nm, and its optical resolution is about **2.0 nm FWHM (full width at half maximum)**. The chip itself has 2048 elements, but the USB-650 reads out 651 points at 1 nm spacing across that range -- you can confirm this from the header of any file it writes. The spectrograph is based on a Czerny-Turner optical design, which has no moving parts.

For this portion of the lab you are welcome to use either your own laptop or one of the much-loved departmental ones. If you use your own you will need to download and install the OceanView control software from Ocean Optics; installers for both Windows and Mac are linked on the Quercus page: OceanView software installation: [link for AST 325](#) / [link for AST 326](#). Make sure you have it installed before the Thursday lab if you want to use your own laptop!

Note that the spectrograph connects to the computer via USB-A (not USB-C), so make sure you have a suitable adapter if you are going to use your own laptop.

3.1.2 Data collection -- what you do in the Thursday lab

1. Plug the USB-650 spectrometer into a USB port **first**, then start the OceanView application. If prompted, select Quick View (not the Wizard or the others). Connect the spectrometer only to a machine you know has OceanView installed.
2. If no device shows up in the data sources window (top left), select Spectrometer / Rescan Devices from the menu indicated by the magnifying-glass icon. If you started the software before plugging in the spectrometer, quit the software, plug in the spectrometer, and try again.
3. In the default operating mode the spectrometer runs in continuous acquisition mode, which, as the name suggests, is like an oscilloscope: the spectrum is continuously scanned at a cadence equal to the integration time.
4. Point the fiber at a hydrogen gas discharge lamp. Make sure you read the following carefully!
 - Do not kink the fiber; bend it only to gentle curvatures that it does not resist.
 - The lamps are very fragile. Handle them carefully.
 - The discharge lamps have a finite lifetime -- the gas is slowly consumed and the electrodes sputter -- and they get hot. Switch a lamp off as soon as you have your spectrum, and never leave one running while you set up something else.
 - The discharge lamps emit ultraviolet light. Do not stare directly into them at close range.
5. The spectrum display is “live” and updates with each new exposure. Wave your hand in front of the entrance aperture and note the change in brightness. The default exposure time is 100 ms, so you should see an immediate response on the plot.
6. Try changing the integration time in the upper-left acquisition group window from the default 100 ms to a longer or shorter time and view the results. Clicking the dot in the bottom-right of the window will display a message indicating whether the spectrum is saturated. Use the set of icons just above the graph to adjust the x- and y-scaling. If your mouse has a scroll wheel, you can use it to zoom in and out.

7. Set the integration time so the brightest line you care about reaches roughly half to three-quarters of the full-scale reading without ever tripping the saturation indicator. A saturated line has a flat top, and its centroid is both biased and much less precise than it looks -- this will quietly wreck your wavelength solution.
8. The easiest way to use the spectrometer is to inspect the live graphical display. This is a very handy option because, for example, it lets you see immediately whether the light source is bright enough to yield useful data.
9. Before taking your spectrum of the hydrogen lamp, find a stable position in which to hold the fiber pointed at the lamp. Note how small movements of the fiber can significantly alter the intensity of the spectrum.
10. When you are happy with the exposure time and the other details of the measurement, save your data (Figure 1). To configure the saving controls, click the gear-with-a-wrench icon labeled "Configure graph saving." To save the spectrum, click the download button (an arrow pointing down into a rectangle). Click "Browse" to select the path, then type a file name in the dialog box.

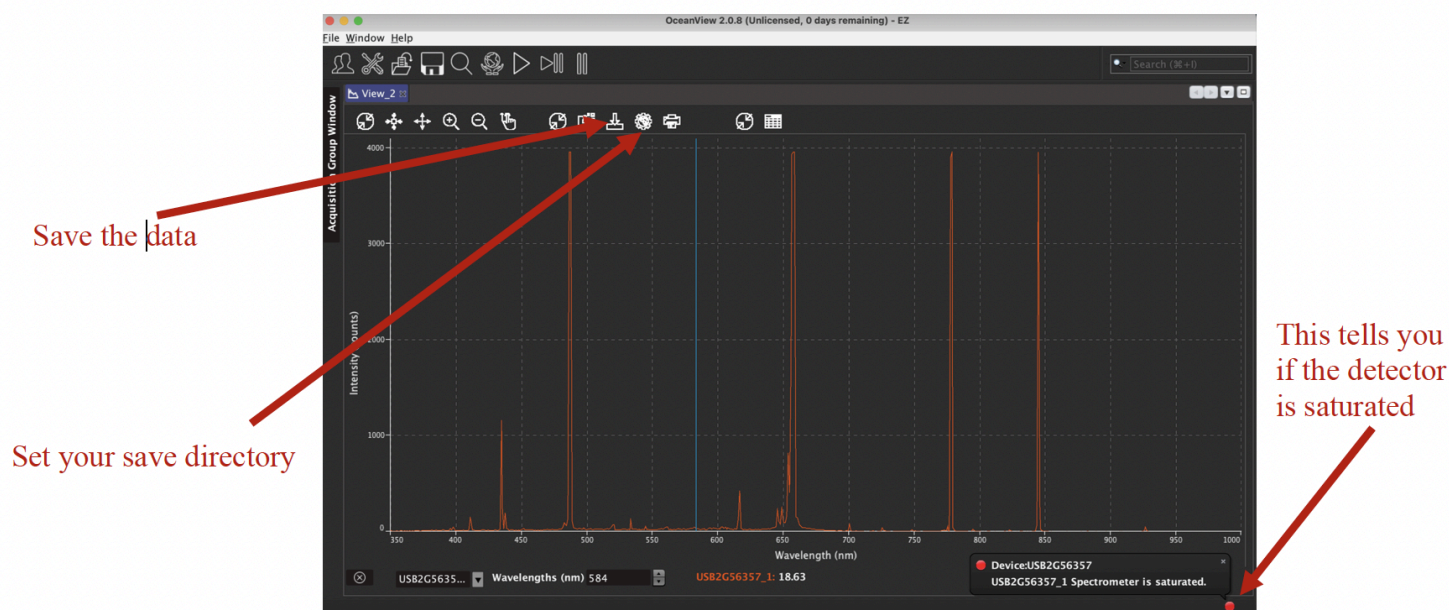


Figure 1. The OceanView window, with the saving controls and the saturation indicator marked. Note that the spectrum shown here is saturated: the four flat-topped lines pinned at 4000 counts are exactly what you must avoid.

11. Take multiple consecutive spectra to combine in your analysis. This will improve your signal-to-noise ratio and reveal fainter lines.
12. A few tips for the saving window:
 - Set the target directory and the file-naming convention here. To keep different observations organized (hydrogen lamp, mystery lamp, ...), either use a "BaseName", which prepends a prefix to the filename, or save to separate directories.

13. Take an ambient spectrum, holding the fiber to the lamp as before but with the lamp switched off. This will let you identify lines associated with interfering sources such as the room lights.
14. **There are two arc lamps containing unknown gases. Take spectra of both** while you have the spectrometer set up -- each takes a minute once your exposure time is right, and you will not get a second chance -- with an ambient spectrum for each, at the same integration time. Write down the tube number printed on each lamp and quote it in your report.
15. Also record spectra of an incandescent lamp, a fluorescent strip light, and (if you can get the fiber to a window) sunlight. You will be asked to compare continuum and line sources in your report, and these take a minute each.

3.1.3 Calibration

1. The spectrometer comes factory-calibrated, allowing it to convert CCD pixel coordinates to wavelength values. However, this calibration is prone to shifting over time. For this part of the lab, perform your own calibration based on measured line positions.
2. The hydrogen gas discharge lamp is filled with partially ionized low-density hydrogen that radiates at known wavelengths. Which series of hydrogen lines can this instrument reach, and which can it not? Look up the wavelengths in the [NIST Atomic Spectra Database](#), and note whether the values you take are air or vacuum wavelengths -- you are making a choice, and it should be a deliberate one.
3. Look at what is in the file. Open one file in a text editor before you write any code: there is a header block, then the data. You may read it programmatically rather than by hand-editing files -- you will have dozens of them. The data contains two columns: work out what each one is, and what the header tells you about how the data have already been processed. Your pixel number is the row index -- the first column is OceanView's own idea of the wavelength, and deciding how much to trust it is part of this exercise.
4. Write a Python script to read in your hydrogen-lamp spectra. Average your N consecutive hydrogen scans to improve the signal-to-noise ratio.
5. Write a Python script to find the centroids (pixel positions) of the brightest features of your spectrum. (Is the location of the maximum pixel enough for getting the centroids?) You do not have to write a peak-finder or a fitting routine from scratch: built-in functions such as `scipy.signal.find_peaks` and `scipy.optimize.curve_fit` are fine, as long as you understand what they compute and say why the quantity you report is the centroid you want.
6. Plot the known wavelengths of the lines you identified as a function of the pixel positions of their centroids.
7. Perform a linear fit to determine the relationship between pixel number and wavelength.
8. Plot the residuals of this linear fit. Comment on any structure you see. Is a linear relation adequate? Can you do better?
9. Using the same principles, perform a higher-order polynomial fit (quadratic, cubic, ...). You only need to try one.

10. Plot the residuals of the higher-order fit and compare them with the linear case.

3.1.4 Plotting spectra

1. Use your fit to plot the raw counts from the spectrograph against wavelength. This is the spectrum of the hydrogen arc lamp.
2. Comment on any other features you see. Can you see any of the fainter hydrogen lines?
3. Compare your hydrogen spectrum with the incandescent lamp, the fluorescent strip light, and sunlight. Which are continuum sources and which are line sources? The fluorescent light is a useful intermediate case -- explain what you see in it.

3.1.5 Mystery gas

1. You took spectra of two unknown lamps. For your report you may identify one of them or both. Identifying both is not required, but a second identification done well, with the candidates argued against each other, is worth more than one.
2. Apply your hydrogen wavelength solution to the unknown spectrum, plot it against wavelength, and use the strongest lines to identify the element. **The gas is one of the five elements listed in Appendix B at the end of this handout**, so this is a multiple-choice question, not a search of the whole periodic table.
3. Identify on the pattern, not on a single coincidence, and think about what your instrument can and cannot see before you count a missing line as evidence.
4. Justify the identification quantitatively: list your measured line wavelengths with uncertainties next to the catalog wavelengths of your candidate element, and say why you rejected the runners-up. "It looks like neon" is not an answer.
5. **Now turn the identification back on the calibration.** Once you know the element, you know the true wavelength of every line you measured in it. Compute the residual -- true wavelength minus the wavelength your hydrogen solution gives -- for each identified line, and plot the residuals against pixel number. How good is the hydrogen solution where you actually used it? Hydrogen gave you four lines, all between 410 and 656 nm; most of the mystery-gas lines fall where hydrogen gave you nothing.
6. Then refit, using the hydrogen lines and the identified mystery-gas lines together. You now have a dozen or more points spread across the whole range instead of four bunched at one end. Show what that does to the residuals and to the fitted coefficients, and revisit the question of fit order with the extra points in hand: with four points a cubic will always look better than a line whether or not the extra terms mean anything; with fifteen you can actually tell. State which solution you adopt and quote its RMS residual in nm -- that is the uncertainty of your wavelength scale.

3.2 Nighttime Observing with the LISA spectrograph

3.2.1 Overview and preparation -- **read before your observatory visit**

From 1-D to 2-D. Everything in Section 3.1 assumed that a spectrum is a list of numbers, because the USB-650 has a single row of pixels. A slit spectrograph on a telescope does not: its detector is a 2-D array, one axis carrying wavelength and the other position along the slit, and what it records is an image (Figure 2).

The LISA spectrograph is dedicated to the spectroscopy of faint objects such as comets, novae, supernovae, faint variable stars, the redshifts of distant galaxies and quasars, and the confirmation of planetary nebulae. Its optical design has been specially optimized for this type of object, with custom-made optics. Its integrated calibration module contains an argon/neon lamp, which provides a large number of emission lines across the LISA spectral range, and a tungsten (halogen) lamp for flat fields. LISA is a slit spectrograph with a resolving power of 600-1000 depending on the slit and configuration.

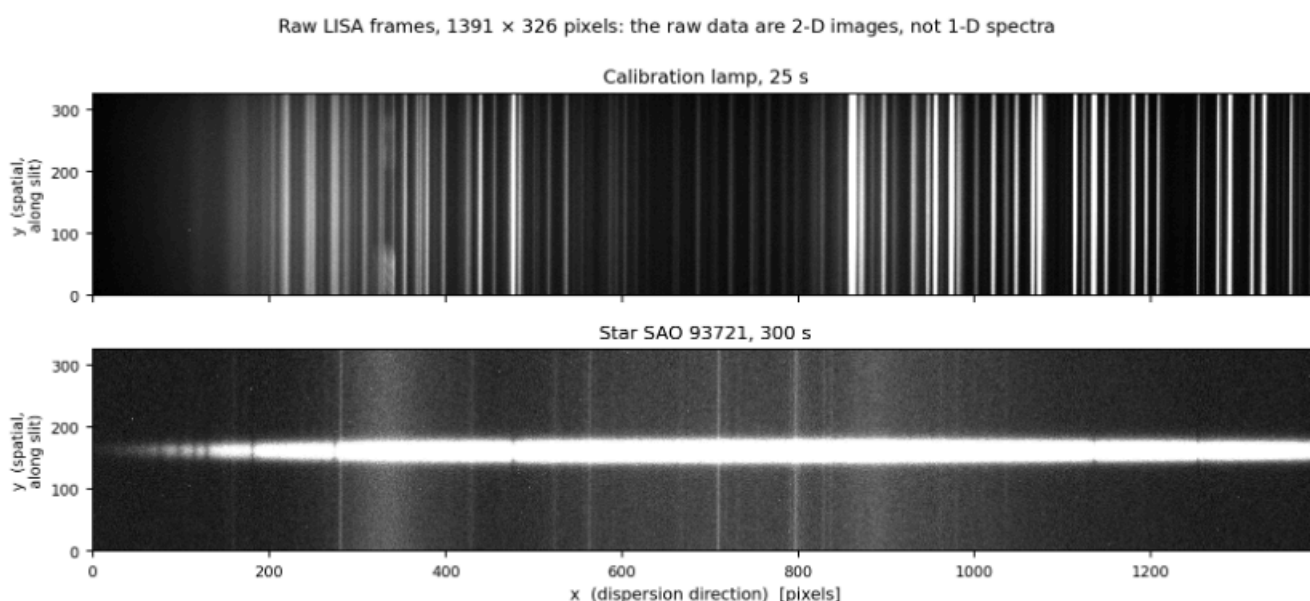


Figure 2. Raw LISA frames from a previous observing run. Top: the argon/neon calibration lamp, a picket fence of emission lines running the full height of the slit. Bottom: a 300 s exposure of a star -- the horizontal streak is the stellar continuum, and the faint vertical lines crossing it are night-sky emission. Neither frame is a spectrum yet: dispersion runs left-right, position along the slit runs up-down, and nothing in either image is labeled in wavelength.

Choosing your own targets

You will not be handed a target list on the night. Your group chooses, and the telescope is pointed where you ask. Come with a written, ranked plan worked out in advance, and with more targets than you expect to need. Groups that arrive without one spend most of a clear night deciding what to look at. In your report, put the plan next to what you actually got, and say for each target whether your expectation was right.

The lists to choose from are in **Appendix A**: Table A1 is a long list of stars brighter than $V = 6$, arranged from the hottest spectral types to the coolest; Table A2 is a shorter list of Solar-System objects, extended objects, emission-line stars, variables and a few other stars that are not ordinary. **Neither list**

has been filtered for you. Some of the stars in Table A1 are below the horizon all night in October, some never rise from Toronto at all, and some are up but too low to be worth observing. Deciding which is which, for every object you consider, is part of the assignment.

What the observing run is for. Three things, and your plan should show all three:

1. **A temperature sequence.** Choose three stars or more that span the spectral sequence -- at minimum one hot star (O, B or A), one intermediate (F, G or K), and one cool (M). The point is to see for yourself how a stellar spectrum changes with temperature: which lines appear, which disappear, and where the light goes. Table A1 gives the spectral type.
2. **Observability and exposure.** For every candidate, work out whether it is up during your slot at a useful altitude, given the date and the latitude of Toronto, and choose an exposure time from its V magnitude.
3. **Something that is not an ordinary star.** From Table A2, plan for one or two objects if they are available -- a planet, the Moon, a nebula, a galaxy, an emission-line star, a variable, a double star or a carbon star -- and reduce and plot them alongside your stellar sequence. Do they look like a stellar photosphere and why?

Is it up?

A target is highest in the sky when the local sidereal time (LST) equals its right ascension. Its altitude a at any other moment follows from

$$\sin a = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos H, \quad H = \text{LST} - \alpha$$

where $\varphi = +43.66^\circ$ is the latitude of the campus observatory, δ and α are the target's declination and right ascension, and H is the hour angle. Three consequences worth internalizing: anything with $\delta \geq +46^\circ$ never sets from Toronto; anything with $\delta \leq -16^\circ$ never gets more than 30° above the horizon; and anything with $\delta < -46^\circ$ never rises at all. Table A1 contains stars in every one of those categories.

Insist on an altitude above about 30° for everything you plan to observe. Below that you are looking through airmass > 2 , so the target is dimmed and reddened; atmospheric dispersion smears the light into a little spectrum of its own before it even reaches the slit, which costs you blue photons; and in practice the surrounding buildings will simply be in the way.

The telescope mount does not allow for a target at an altitude of $80\text{--}90^\circ$, so make sure you do not select a target in this range during your observing slot.

Is it bright enough, and how long do you expose?

This year the spectrograph is on the campus **10-inch** telescope. Aim for a signal-to-noise ratio of about 100 per resolution element or higher in the yellow, and keep every exposure at or under 300 s. The actual exposure time depends on the altitude of the target. But usually for the 10-inch telescope with LISA spectrograph, 10-30 s exposure gives enough signal-to-noise ratio for a $V = 0$ star based on previous observations. From this anchor and the way signal scales with magnitude and exposure time (remember 1 magnitude larger = $2.512\times$ fainter), work out the exposure you need for each of your targets, and put it in your plan. Then look at the first frame of each target on the night and adjust: the

saturation level is 65 535 ADU (analog-to-digital units), and you want to peak below 40 000 to avoid saturation. Usually a count of 2000 ADU per pixel should give enough signal-to-noise ratio (and going higher does no harm if the exposure is short).

Plan your 1.5 hr session wisely

Budget roughly 10-15 minutes per target once slewing, acquiring it on the slit, exposing and saving the slit-view image are counted, along with the calibration frames (e.g. flat field, arc, dark) and remember that every exposure (e.g. flat, arc, science) commits you to darks of the same length: a 25 s arc frame, for example, needs three 25 s darks, three so that you can combine them and reduce the noise. Therefore, it is a good idea to take the same science exposure time for targets with similar brightness. Plan to observe four to seven targets together with the calibration frames for your 1.5 hr session.

What you should expect to see

One caution before the list: until you have corrected for the response of the instrument, the raw slope of every spectrum is set by the instrument, not the star. The differences between classes are in the lines and bands.

- **O and B (hotter than ~10 000 K):** a smooth continuum with relatively few lines -- hydrogen Balmer absorption and, in the hotter ones, neutral helium.
- **A (~7500-10 000 K):** Balmer absorption at its strongest and broadest.
- **F and G (~5200-7500 K):** Balmer lines weakening and the G band appearing near 4300 Å. Ca II H and K (3934, 3968 Å) are the classic diagnostic, but they sit right at the blue edge of LISA's range where the throughput is very low, so do not count on them.
- **K (~3700-5200 K):** the strong, broad features survive at this resolution: Mg b near 5175 Å, the Na D pair at 5890/5896 Å, Ca I 4227 Å and the G band. In the later K types you will start to see the first TiO bands.
- **M (~2400-3700 K):** broad TiO absorption bands that chop the continuum into a sawtooth. Unmistakable, and unlike anything in a hotter star.
- **In every spectrum:** telluric absorption from the Earth's atmosphere -- the O₂ B band at 6867 Å, and water-vapor bands beyond about 7000 Å. (The much stronger O₂ A band at 7600 Å lies just past LISA's red edge, so you will not see it.) These features do not belong to the star, they do not shift with it, and they get deeper as the airmass rises. Being able to say which features in your spectrum are stellar, which are telluric and which are instrumental is one of the things this lab is for.

3.2.2 Data collection -- what you do at the telescope

1. The observing sessions take place at the campus observatory's 10-inch telescope, on the 15th floor of McLennan Physical Laboratories, 60 St. George Street. Take the elevator to the 14th floor. Turn right when you leave the elevator, walk to the end of the hall, and turn right again. Halfway down the hall there is a staircase on your right. Take the stairs up and meet the telescope support staff on the roof.

2. Arrive at your group's designated time slot with a USB thumb drive to take your data away. Michael Williams (williams@astro.utoronto.ca) or Ethen Sun (ethen.sun@mail.utoronto.ca), who look after the lab equipment and operate the telescope, will be there to set up and to help you collect your data.
3. Dress for it. The dome is not heated and is open to the outside air; on an October or November night that means whatever it is outside, plus wind. Bring a hat and gloves.
4. Point the telescope at several astronomical targets (4-7) and record a frame for each. Observe the targets you planned in Section 3.2.1, in priority order, and re-check the altitudes against what you predicted.
5. Record a set of calibration frames: an image of the argon/neon lamp (for the wavelength calibration), images of the tungsten lamp (for the flat field), and a set of dark frames (for dark subtraction).
6. Record in a table in your final report the targets, UT dates and times, exposure times, sky conditions, and who was in attendance. Compare this with your original plan.

3.2.3 Detector calibration

1. Load all of your dark images into Python and median-combine them, at each exposure time, to create a master dark.
2. Do the same for your flats: median-combine the tungsten-lamp frames, subtract the master dark of the matching exposure time, and divide the result by its own median, so that the flat is normalized to 1. Then, for each target frame, apply

$$P_i = (R_i - D_i) / F_i^{\text{norm}}$$

where P_i is the i^{th} pixel of the calibrated image, R_i the raw target frame, D_i the master dark of the same exposure time, and F_i^{norm} the master flat normalized to a median of 1.

What this does, and what it does not. The flat frame is the lamp's spectrum, times the instrument's throughput, times each pixel's own sensitivity. Dividing your star by it removes the pixel-to-pixel sensitivity and cancels the instrument throughput -- but it also divides the star by the lamp's spectrum. The lamp is a halogen bulb, much redder than any star you will observe, so every spectrum you produce this way is tilted toward the blue by the same factor. The shapes are therefore correct relative to each other, which is what you need to compare spectral types, but not correct in absolute terms. In professional work the flat is first divided by a smooth fit to its own shape, so that only the pixel-to-pixel term is left, and the instrument throughput is measured separately with a star of known spectrum; we will discuss this in class. One practical consequence of the simple method: where the lamp gave almost no counts, the division amplifies noise. Look at your flat, decide where to cut, and say so.

What this part of the calibration does: subtracting the master dark removes the fixed electronic offset, the accumulated dark current and most of the bad pixels. Dividing by the flat corrects for pixel-to-pixel sensitivity differences and cancels the instrument response, at the price of imprinting the lamp's own spectrum.

3.2.4 Extracting a 1-D spectrum

Your calibrated frame is still an image. To get a spectrum out of it:

1. Collapse the frame along the dispersion direction to get the spatial profile of the target across the slit. Measure its width; it is wider than you might guess.
2. Find the trace -- the center of that profile as a function of column -- and check whether it is flat. Do not assume.
3. Sum the counts in an aperture centered on the trace. Choosing its width is a signal-to-noise trade-off; choose, and justify.
4. Estimate the sky from rows on either side of the trace (the median), and subtract it scaled to the number of rows in your aperture. The sky lines in Figure 2 should largely disappear if you have done this right; whatever remains is real. Your sky rows are further from the trace than your aperture is wide, so if the lines are tilted the sky lines there are slightly shifted relative to your target rows -- work out how big that shift is for your frame before you decide whether it matters.
5. Extract the wavelength calibration lamp frame with exactly the same rows and the same sum. This is the spectrum you calibrate in 3.2.5.

3.2.5 Wavelength calibration

1. A wavelength calibration needs to be performed, like the one you did for the Ocean Optics spectrograph. It is slightly more involved here because this is a 2-dimensional detector rather than the 1-dimensional detector used by the Ocean Optics unit.
2. First plot the image of the calibration frame. You should see a series of lines running across the image.
3. **The lamp is argon/neon:** if you match against a neutral-neon list alone you will fail on the tallest lines in your own data and have no anchors in the blue. Pull Ne I, Ar I and Ar II from NIST, and identify on the pattern.
4. Check whether the lines are tilted with respect to the detector axes. (similar to Section 3.2.4)
5. The wavelength calibration is done on the 1-D lamp spectrum you extracted in 3.2.4 with the same rows and the same sum as the star, so that the lamp spectrum and the science spectrum are sampled identically. Subtract the matched dark from the lamp frame as you did for the star; do not flat-field it -- the calibration uses line positions, which the flat does not change.
6. Compare the line centroids in that extracted lamp spectrum with the known wavelengths of the brightest neon and argon transitions, just as you did with hydrogen. Plot wavelength against centroid position and perform a polynomial fit. There are far more lines here than in the hydrogen lamp, so be selective: isolated, unsaturated lines spread across the whole range.
7. Quote the uncertainty of your wavelength solution, report the dispersion in Å per pixel and the wavelength range you cover.

8. Only now apply that solution to your sky-subtracted science spectrum from 3.2.4, and plot it against wavelength. Everything stays in pixel units until this step. The solution is valid for spectra extracted from the same rows, and only there: a spectrum extracted from a different part of the slit needs its own lamp extraction if the lines are tilted.

3.2.6 Spectral analysis

1. Using both your wavelength calibration and your detector calibration, plot the spectrum of each of your astronomical targets.
2. Compare and comment on the most important features.
 - Concretely: identify the Balmer absorption lines in the hot stars and compare your measured wavelengths with the laboratory values you used in Section 3.1. If you observed the Moon or a planet, describe what its spectrum looks like, and explain what you see and why. Say which features come from the star, which from the Earth's atmosphere, and which from the instrument -- and how you can tell. Your sky spectrum, from downtown Toronto, is a good place to start: identify what is in it, and then look for the natural airglow line [O I] 5577 Å and say what its presence or absence tells you.
3. Estimate the resolving power $R = \lambda/\Delta\lambda$ you actually achieved, from the width of isolated calibration lines at several different wavelengths, and compare with the manufacturer's quoted 600-1000. It is not a single number: R varies substantially across the frame, and saying by how much, and in which direction, is part of characterizing the instrument.

4. What to hand in

One report per student, even though the data are taken in groups. It should stand on its own: someone who has not read this handout should be able to follow what you did and why. At a minimum it should contain:

- Your wavelength solution for the Ocean Optics spectrometer, with residual plots for at least two fit orders, and a statement of which you adopted and why.
- The identification of one or both mystery gases (and the tube numbers), with measured versus catalog wavelengths and uncertainties and the reasons the other candidates were rejected; the residuals of your hydrogen solution on the identified lines; and the refitted solution.
- Your observing plan and the log of what you actually got: targets, UT dates and times, exposure times, observers, conditions, and one sentence per target on why you chose it, what you expected, and whether you were right.
- At least one reduced, wavelength-calibrated, extracted spectrum per target, with the features you discuss marked on the plot, together with your LISA wavelength solution (its RMS, dispersion and range) and the resolving power you measured.
- A short honest account of what went wrong, since something will. Negative results reported clearly are worth more marks than a result that has been quietly tidied up.

Format and marking. Your report is marked with the rubric in Section 5.6 of the course syllabus, which also sets the format rules (page limit, font size, title block). Refer to it before you write, and use its section headings: Abstract, Introduction, Observations & Data, Data reduction & Methods, Data analysis & Modeling, Discussion, and an Appendix presenting your own code. The page limit is tight for this lab, so construct your figures wisely, e.g. the stellar spectra from the campus observatory can go in one plot, either with multi-panels, or overplotted with a vertical offset and labelled by spectral type.

Where the pieces above belong. The observing plan, the log table and your spectrometer settings go in Observations & Data. The dark and flat corrections, the extraction and sky subtraction, and both wavelength solutions with their residual plots go in Data reduction & Methods. The line identifications, the mystery-gas argument, the temperature sequence, the comparison of stellar and non-stellar spectra and the resolving power you measured go in Data analysis & Modeling. What you expected against what you got, what went wrong and why, and what you would do differently go in the Discussion. The essential pieces of code you wrote yourselves, each with a short description, go in the Appendix, and the main text should point to them.

Appendix A: Target lists

Use these with Section 3.2. Nothing here has been filtered for observability from Toronto or for the date: that is your job. All positions are J2000.

Table A1. Stars brighter than $V = 6.0$, arranged by spectral class from hottest to coolest, with a dozen or more per class spread around the sky in right ascension, including the brightest evening targets for October. Within each class they are in order of right ascension. Coordinates, magnitudes and spectral types are from the Bright Star Catalogue (Hoffleit & Warren 1991, VizieR V/50); the temperature ranges are approximate and are given so that you can place each star on a temperature axis.

Star	RA (J2000)	Dec (J2000)	V	Sp. type	Note
O: hotter than ~30 000 K					
46 ξ Per -- Menkib	03 58 57.9	+35 47 28	4.04	O7.5III(n)((f))	
9 α Cam	04 54 03.0	+66 20 34	4.29	O9.5Ia	
34 δ Ori -- Mintaka	05 32 00.4	-00 17 57	2.23	O9.5II	
50 ζ Ori -- Alnitak	05 40 45.5	-01 56 34	2.05	O9.7Ib	
ζ Pup -- Naos	08 03 35.1	-40 00 12	2.25	O5f	
HR 4908	12 55 57.0	-56 50 10	5.32	O9Ib	
δ Cir	15 16 56.7	-60 57 27	5.09	O8.5V	
13 ζ Oph	16 37 09.5	-10 34 02	2.56	O9.5Vn	
HR 6245	16 51 33.7	-41 13 50	5.22	O8Iaf	
68 Cyg	21 18 27.2	+43 56 45	5.00	O7.5III:n((f))	
14 Cep	22 02 04.6	+58 00 02	5.56	O9Vn	
19 Cep	22 05 08.9	+62 16 48	5.11	O9.5Ib	
22 λ Cep	22 11 30.7	+59 24 52	5.04	O6I(n)fp	
10 Lac	22 39 15.7	+39 03 01	4.88	O9V	
B: ~10 000-30 000 K					
21 α And -- Alpheratz	00 08 23.3	+29 05 26	2.06	B8IVpMnHg	
88 γ Peg -- Algenib	00 13 14.2	+15 11 01	2.83	B2IV	
α Eri -- Achernar	01 37 42.9	-57 14 12	0.46	B3Vpe	<i>emission lines</i>
26 β Per -- Algol	03 08 10.1	+40 57 20	2.12	B8V	<i>eclipsing binary; check its phase</i>
19 β Ori -- Rigel	05 14 32.3	-08 12 06	0.12	B8Ia:	
21 ϵ CMa -- Adhara	06 58 37.5	-28 58 20	1.50	B2II	
κ Vel -- Markeb	09 22 06.8	-55 00 39	2.50	B2IV-V	
32 α Leo -- Regulus	10 08 22.3	+11 58 02	1.35	B7V	
67 α Vir -- Spica	13 25 11.6	-11 09 41	0.98	B1III-IV+B2V	<i>composite spectrum (binary)</i>
β Cen -- Hadar	14 03 49.4	-60 22 23	0.61	B1III	

35 λ Sco -- Shaula	17 33 36.5	-37 06 14	1.63	B2IV+B	<i>composite spectrum (binary)</i>
20 ϵ Sgr -- Kaus Australis	18 24 10.3	-34 23 05	1.85	B9.5III	
18 δ Cyg -- Fawaris	19 44 58.5	+45 07 51	2.87	B9.5IV+F1V	<i>composite spectrum (binary)</i>
α Pav -- Peacock	20 25 38.9	-56 44 06	1.94	B2IV	
α Gru -- Alnair	22 08 14.0	-46 57 40	1.74	B7IV	
54 α Peg -- Markab	23 04 45.7	+15 12 19	2.49	B9V	
A: ~7500-10 000 K					
37 δ Cas -- Ruchbah	01 25 49.0	+60 14 07	2.68	A5III-IV	
6 β Ari -- Sheratan	01 54 38.4	+20 48 29	2.64	A5V	
4 β Tri	02 09 32.6	+34 59 14	3.00	A5III	
34 β Aur -- Menkalinan	05 59 31.7	+44 56 51	1.90	A2IV	
9 α CMa -- Sirius	06 45 08.9	-16 42 58	-1.46	A1Vm	
δ Vel -- Alsephina	08 44 42.2	-54 42 30	1.96	A1V	
β Car -- Miaplacidus	09 13 12.0	-69 43 02	1.68	A2IV	
γ Cen	12 41 31.0	-48 57 35	2.17	A1IV	
77 ϵ UMa -- Alioth	12 54 01.7	+55 57 35	1.77	A0pCr	
55 α Oph -- Rasalhague	17 34 56.1	+12 33 36	2.08	A5III	
3 α Lyr -- Vega	18 36 56.3	+38 47 01	0.03	A0Va	
17 ζ Aql -- Okab	19 05 24.6	+13 51 48	2.99	A0Vn	
53 α Aql -- Altair	19 50 47.0	+08 52 06	0.77	A7V	
50 α Cyg -- Deneb	20 41 25.9	+45 16 49	1.25	A2Ia	
5 α Cep -- Alderamin	21 18 34.8	+62 35 08	2.44	A7V	
24 α PsA -- Fomalhaut	22 57 39.1	-29 37 20	1.16	A3V	
F: ~6000-7500 K					
11 β Cas -- Caph	00 09 10.7	+59 08 59	2.27	F2III-IV	
24 η Cas -- Achird	00 49 06.0	+57 48 57	3.44	F9V+dM0	<i>F9V primary of a visual binary</i>
2 α Tri -- Mothallah	01 53 04.9	+29 34 44	3.41	F6IV	
1 α UMi -- Polaris	02 31 48.7	+89 15 51	2.02	F7:Ib-II	<i>Cepheid variable</i>
33 α Per -- Mirfak	03 24 19.4	+49 51 40	1.79	F5Ib	
α Car -- Canopus	06 23 57.1	-52 41 45	-0.72	F0II	
10 α CMi -- Procyon	07 39 18.1	+05 13 30	0.38	F5IV-V	
15 ρ Pup -- Tureis	08 07 32.6	-24 18 15	2.81	F6IIpDel Del	
25 θ UMa	09 32 51.4	+51 40 38	3.17	F6IV	
29 γ Vir -- Porrima	12 41 39.6	-01 26 58	3.65	F0V	
β TrA	15 55 08.5	-63 25 50	2.85	F2III	

θ Sco -- Sargas	17 37 19.2	-42 59 52	1.87	F1II	
41 π Sgr -- Albaldah	19 09 45.8	-21 01 25	2.89	F2II	
30 δ Aql	19 25 29.9	+03 06 53	3.36	F3IV	
37 γ Cyg -- Sadr	20 22 13.7	+40 15 24	2.20	F8Ib	
G: ~5200-6000 K					
β Hyi	00 25 45.1	-77 15 15	2.80	G2IV	
16 β Cet -- Diphda	00 43 35.4	-17 59 12	2.04	G9.5IIICH-1	
23 γ Per	03 04 47.8	+53 30 23	2.93	G8III+A2V	<i>composite spectrum (binary)</i>
13 α Aur -- Capella	05 16 41.4	+45 59 53	0.08	G5IIIe+G0III	<i>spectroscopic binary, two giants</i>
9 β Lep -- Nihal	05 28 14.7	-20 45 34	2.84	G5II	
17 ϵ Leo	09 45 51.1	+23 46 27	2.98	G1II	
μ Vel	10 46 46.2	-49 25 12	2.69	G5III+G2V	<i>composite spectrum (binary)</i>
9 β Crv -- Kraz	12 34 23.2	-23 23 48	2.65	G5II	
α^1 Cen -- Rigil Kentaurus	14 39 35.9	-60 50 07	-0.01	G2V	
14 η Dra -- Athebyne	16 23 59.5	+61 30 51	2.74	G8-IIIab	
27 β Her -- Kornephoros	16 30 13.2	+21 29 23	2.77	G7IIIa	
40 ζ Her	16 41 17.2	+31 36 11	2.81	G0IV	
23 β Dra -- Rastaban	17 30 26.0	+52 18 05	2.79	G2Ib-IIa	
22 β Aqr -- Sadalsuud	21 31 33.5	-05 34 16	2.91	G0Ib	
34 α Aqr -- Sadalmelik	22 05 47.0	-00 19 11	2.96	G2Ib	
44 η Peg -- Matar	22 43 00.1	+30 13 17	2.94	G2II-III+F0V	<i>composite spectrum (binary)</i>
K: ~3700-5200 K					
18 α Cas -- Schedar	00 40 30.5	+56 32 14	2.23	K0IIIa	
13 α Ari -- Hamal	02 07 10.4	+23 27 45	2.00	K2-IIICa-1	
87 α Tau -- Aldebaran	04 35 55.2	+16 30 33	0.85	K5+III	
78 β Gem -- Pollux	07 45 18.9	+28 01 34	1.14	K0IIIb	
ϵ Car -- Avior	08 22 30.8	-59 30 35	1.86	K3III+B2:V	<i>composite spectrum (binary)</i>
50 α UMa -- Dubhe	11 03 43.7	+61 45 03	1.79	K0IIIa	
16 α Boo -- Arcturus	14 15 39.7	+19 10 57	-0.04	K1.5IIIFe-0.5	
α^2 Cen -- Toliman	14 39 36.1	-60 50 08	1.33	K1V	
7 β UMi -- Kochab	14 50 42.3	+74 09 20	2.08	K4-III	
α TrA -- Atria	16 48 39.9	-69 01 40	1.92	K2IIb-IIIa	
67 π Her	17 15 02.8	+36 48 33	3.16	K3IIab	
60 β Oph -- Cebalrai	17 43 28.4	+04 34 02	2.77	K2III	

33 γ Dra -- Eltanin	17 56 36.4	+51 29 20	2.23	K5III	
50 γ Aql -- Tarazed	19 46 15.6	+10 36 48	2.72	K3II	
53 ϵ Cyg -- Aljanah	20 46 12.7	+33 58 13	2.46	K0-III	
8 ϵ Peg -- Enif	21 44 11.2	+09 52 30	2.39	K2Ib	
M: ~2400-3700 K					
43 β And -- Mirach	01 09 43.9	+35 37 14	2.06	M0+IIIa	
92 α Cet -- Menkar	03 02 16.8	+04 05 23	2.53	M1.5IIIa	
25 ρ Per -- Gorgonea Tertia	03 05 10.6	+38 50 25	3.39	M4II	<i>semi-regular variable</i>
58 α Ori -- Betelgeuse	05 55 10.3	+07 24 25	0.50	M1-2Ia-Iab	
13 μ Gem -- Tejat	06 22 57.6	+22 30 49	2.88	M3IIIab	
34 μ UMa -- Tania Australis	10 22 19.7	+41 29 58	3.05	M0III	
1 λ Dra -- Giasar	11 31 24.2	+69 19 52	3.84	M0IIICa-1	
γ Cru -- Gacrux	12 31 09.9	-57 06 48	1.63	M3.5III	
20 σ Lib -- Brachium	15 04 04.2	-25 16 55	3.29	M3-III	
1 δ Oph -- Yed Prior	16 14 20.7	-03 41 40	2.74	M0.5III	
21 α Sco -- Antares	16 29 24.4	-26 25 55	0.96	M1.5Iab-Ib+B4Ve	<i>composite spectrum (binary)</i>
64 α^1 Her -- Rasalgethi	17 14 38.9	+14 23 25	3.48	M5Ib-II	<i>semi-regular variable; bright companion 4.9 arcsec away</i>
12 γ Sge	19 58 45.4	+19 29 32	3.47	M0-III	
β Gru -- Tiaki	22 42 40.1	-46 53 05	2.10	M5III	
73 λ Aqr	22 52 36.9	-07 34 47	3.74	M2.5IIaFe-1	
53 β Peg -- Scheat	23 03 46.5	+28 04 58	2.42	M2.5II-III	

Table A2. Objects that are not ordinary stars. The Moon and planets move, so no coordinates are given: find their positions for your date. Coordinates for the fixed objects are from SIMBAD.

Object	RA (J2000)	Dec (J2000)	V	Type / spectrum	Note
Moon	moves	moves	-13 to -5	Solar-System object	<i>needs only seconds; brightens the whole sky when near full</i>
Jupiter	moves	moves	≈ -2.0	Solar-System object	
Saturn	moves	moves	$\approx +0.4$	Solar-System object	
Uranus	moves	moves	$\approx +5.6$	Solar-System object	

Neptune	moves	moves	$\approx +7.7$	Solar-System object	
M 31 -- Andromeda Galaxy	00 42 44.3	+41 16 08	3.4 (integrated)	galaxy; absorption-line spectrum of old stars	<i>extended and faint per unit area</i>
M 42 -- Orion Nebula	05 35 16.8	-05 23 15	extended	H II region	<i>bright emission lines; point at θ^2 Ori A or the Trapezium; the nebula fills the rest of the slit either way</i>
27 γ Cas	00 56 42.5	+60 43 00	2.39	B0.5IVpe	<i>Hα in emission (Be star)</i>
123 ζ Tau -- Tianguan	05 37 38.7	+21 08 33	3.03	B1IVe shell	<i>Be shell star; emission varies</i>
10 β Lyr -- Sheliak	18 50 04.8	+33 21 46	3.42	B8.5Ib-II	<i>eclipsing binary, emission lines</i>
34 P Cyg -- P Cygni	20 17 47.2	+38 01 59	4.82	B1-2Ia-0ep	<i>classic P Cygni profiles</i>
68 o Cet -- Mira	02 19 20.8	-02 58 40	2-10 var.	M5/6IIIe	<i>Mira variable; check its phase first</i>
6 β Cyg -- Albireo	19 30 43.3	+27 57 35	3.1 + 5.1	K3II+B9.5V (A); B8Ve (B)	<i>double star 34 arcsec apart: put both components on the slit for two spectra in one frame -- check if the slit is rotatable</i>
27 δ Cep	22 29 10.3	+58 24 55	3.5-4.4 var.	F5Ib-G1Ib	<i>prototype Cepheid, $P = 5.37$ d; spectral type changes with phase, so compare with another group's spectrum</i>
55 η Aql	19 52 28.4	+01 00 20	3.5-4.3 var.	F6Ib-G4Ib	<i>Cepheid, $P = 7.18$ d</i>
1 o And	23 01 55.3	+42 19 34	3.62	B6IV/V shell	<i>Be shell star; Hα emission varies</i>
ϕ Per	01 43 39.6	+50 41 19	4.06	B1.5V:e shell	<i>Be shell star; Hα emission</i>
19 Psc -- TX Psc	23 46 23.5	+03 29 13	≈ 5.0 var.	C-N6 (carbon star)	<i>C₂ Swan bands instead of TiO: the other kind of red star</i>
χ Cyg	19 50 33.9	+32 54 51	3.3-14.2 var.	S6-9/1-2e (S-type Mira)	<i>ZrO bands; check its phase first</i>

Appendix B: The candidate gases

The unknown gas in each of the two arc lamps is one of the following five elements: He, Ne, Ar, Kr, Xe.

All five are noble gases. Discharge tubes are almost always filled with noble gases because they are monatomic, so a tube gives a pure atomic line spectrum with no molecular bands; because they are chemically inert, so nothing reacts with the electrodes or the glass and the tube lasts for years; and because their closed shells give bright, well-separated lines at modest current.

Hydrogen is your calibration lamp, and its lines in the range of this spectrometer are given below. For the five candidates, retrieving the line list is part of the exercise: get it from the NIST Atomic Spectra Database as described here, and be able to say where each wavelength you use came from.

Calibration lamp	Prominent emission lines (air, nm)
H	410.2, 434.0, 486.1, 656.3

Two practical notes. At this spectrometer's resolution, lines closer together than about 2 nm merge into a single feature, so a catalog may list two or three lines where you see one peak. And a used discharge tube is rarely pure: traces of water give an H α line at 656.3 nm in tubes of almost any gas, so a line there is not by itself evidence for hydrogen.

Getting a line list from NIST

1. Go to physics.nist.gov/PhysRefData/ASD/lines_form.html. In **Spectrum** type the species, e.g. Ne I for the neutral atom or Ne I - II for neutral and singly ionized together -- a discharge tube excites both.
2. Set **Lower** = 350 and **Upper** = 1000, and choose **nm** in the wavelength-units menu.
3. Under **Format output** choose **CSV (text)**. Click **Show Advanced Settings**: under *Wavelength Data* tick **Observed**; leave the wavelength-type setting on its default, *Vacuum (< 200 nm)*, *Air (200-2,000 nm)* -- do **not** choose "Vacuum (all wavelengths)"; tick **Relative Intensity**; choose **Only with observed wavelengths or intensities**; set *Display output* to **in its entirety**. Click **Retrieve Data** and save the file.

Two traps. First, air versus vacuum: your spectrometer measures in air, and a vacuum-wavelength list is shifted redward by 0.11 nm at 400 nm rising to 0.27 nm at 1000 nm -- enough to mis-identify lines at the red end. Second, the relative intensities in NIST are compiled from different papers and, in NIST's own words, "have meaning only within a given spectrum": they are not comparable between elements, not between Ne I and Ne II, and not reliably even within one element. Use them as a rough guide, and expect the relative heights in your spectrum to differ from them.

From Python instead. The same query can be made from a script with astroquery, which returns the list as a table with the Excel quoting already removed:

```
from astroquery.nist import Nist
```

```
import astropy.units as u
```

```
t = Nist.query(350*u.nm, 1000*u.nm, linename="Ne I", wavelength_type="vac+air")
```

Use "Ne I-II" for neutral and ionized together, and keep the columns Observed and Rel. NIST's server is sometimes slow or refuses a query; if so, simply run it again.